CULTIVATION GUIDE | SYMPTOMS

Yellowing and Discolouration in Paddy: Telling the Causes Apart

A farmer reporting yellow leaves after heavy rain has several plausible causes, and the right action differs sharply between them.

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Teaching edition for the STEMLink AI Engineer Bootcamp. Content restructured from AgriVibe knowledge base AV-KB-001 Rev 1.1, which grounds its agronomy in Department of Agriculture and RRDI guidance for Sri Lanka. Not for use with real farmers until the verification items in AV-POL-03 are closed.

1. What may and may not be concluded

Yellowing and weak growth may be associated with nitrogen shortage. Pale, generally yellowish-green leaves with reduced growth and reduced tillering are consistent with it. However, in Sri Lankan conditions several other causes produce similar-looking symptoms: excess water and poor drainage, iron toxicity (common in Wet Zone and Intermediate Zone valley fields), salinity arising from poor surface drainage, pest damage, and disease.

Satellite data cannot confirm nitrogen deficiency, and neither can a farmer's description alone. Nitrogen shortage is one possibility among several, and the decision belongs with an inspection and an authoritative source.

2. Nitrogen shortage

Generally pale, yellowish-green leaves across the plant, with reduced growth and reduced tillering, tending to appear more uniformly than water-related problems. The leaf colour chart is the recognised field check. Applying nitrogen is appropriate only once this is established and conditions are suitable.

3. Iron toxicity (bronzing)

Occurs in flooded soils and is found particularly in the rice soils of the up-country and low-country Wet Zone and the Intermediate Zone, commonly in flat valleys where interflow from adjacent highland emerges within the valley. Symptoms are tiny reddish-brown spots beginning on the lower, older leaves, often from the leaf tip, which spread and merge until leaves turn orange-brown and die; roots may be coarse and darkened. It is associated with prolonged flooding and with an inadequate supply of other nutrients. Intercepting the interflow by digging drains at the boundary between the paddy land and the adjacent highland can alleviate it.

> **WHY THIS DIFFERENTIAL MATTERS**
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> Bronzed or reddish-brown discolouration following heavy rain in a Wet Zone or Intermediate Zone field must not be treated as a nitrogen problem. Adding urea is not the remedy for iron toxicity, and a farmer acting on that advice would spend money to no effect while the actual drainage problem persists.

4. Salinity

Can build up where surface drainage is lacking, which is the principal reason for salinity development in Sri Lankan paddy fields. Seepage and runoff collect in depressions or low-lying areas, sometimes because drainage ways have been blocked deliberately or through neglect, and evaporate, leaving salts behind. Improving drainage addresses the problem.

5. Waterlogging and lodging

Excess standing water, physical flattening of the crop by rain or wind, and nutrient loss through runoff over the bunds all follow heavy rainfall, and all reduce measured greenness.

6. Pests and disease

Not diagnosable by this system at all. Any suspicion is routed to the Agriculture Instructor for the area or to the 1920 agricultural advisory service.

7. Questions that distinguish the causes

- Is the colour pale yellow, or reddish-brown and bronzed?
- Are older lower leaves affected first, or the whole plant evenly?
- Is it spread across the field, or concentrated in low or poorly drained parts?
- Is water standing and failing to drain away?
- Does the affected area sit where water enters from adjacent higher ground?